A Market Risk Report in Python: VaR, Backtesting, Stress Tests and Tracking Error

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1 The Core Idea

An asset manager reports the same four things to its risk committee: how much the portfolio can lose in one day, whether that number has been reliable, what happens in a crisis, and how far the portfolio sits from its benchmark. The package computes all four for a five-asset portfolio: 30% EU equities, 30% US equities, 20% government bonds, 15% investment-grade credit and 5% gold.

- **Same data, five answers:** the 99% one-day VaR ranges from 1.28% to 2.15% depending on the method.
- **Synthetic data:** the returns come from a multi-asset GARCH(1,1) (Bollerslev, 1986) with Student-t shocks (1,500 days, seed 42). The numbers show how the methods behave. They are not the risk of a real fund.

2 Why It Matters

- **The method is a choice, and it moves the number.** On one 1,500-day path the five VaRs differ by a factor of 1.7. A committee that sets a limit on one of them is also choosing a method.
- **One year of backtest cannot tell models apart.** With 250 days and 2.5 expected breaches, a perfectly reasonable VaR is rejected by the Kupiec (1995) test in a large share of samples (Section 6). A green traffic light (Basel Committee, 1996) is weak evidence.
- **History is shorter than the risk.** The worst of the 1,500 simulated days is -2.74%. The 2008 replay is -27.65%, ten times larger. Losses are also asymmetric: after a loss d the gain needed is $d/(1-d)$, so -27.65% needs $+38.2\%$.
- **Relative risk is what a client sees.** A fund with a benchmark is judged on the gap to it. The portfolio has a volatility of 12.5% a year but a forecast tracking error of 2.13%, and these are two different limits.

3 Assumptions

Each line states what the code assumes and which way the result moves if it is wrong.

- **Known data process.** Daily returns follow a GARCH(1,1) with $\alpha = 0.08$, $\beta = 0.90$ for all five assets, unconditional volatilities of 18, 20, 6, 9 and 15% and drifts of 6, 7, 1, 3 and

2% a year. Shocks are Student-t with 6 degrees of freedom, correlated through one fixed matrix, and the five assets share one chi-square draw per day. Real assets have jumps and crisis correlations, so real tails would be heavier. The dates (business days from 2 January 2019) are labels only.

- **Linear positions, fixed weights.** The portfolio return is $w^\top r$ with daily rebalancing, no costs and no liquidity horizon. Options would make the loss non-linear in the shocks.
- **One day, mean kept.** The normal and Cornish-Fisher (Cornish and Fisher, 1938) VaRs are centred on the sample mean (0.030% a day). The historical VaR is the raw empirical quantile, so the drift stays in the returns. The Euler decomposition has no mean term. A positive drift lowers the first two by about 0.03 points here, and more at longer horizons.
- **Normal model.** The normal VaR uses the sample standard deviation and a symmetric tail. With excess kurtosis of 4.8 in the sample it is too low in the tail.
- **Cornish-Fisher.** The expansion uses the sample skewness and kurtosis, and is reliable only for moderate values of both. One outlier in a short sample moves it a lot.
- **Age-weighting decay.** The BRW VaR (Boudoukh, Richardson and Whitelaw, 1998) uses $\lambda = 0.94$, the daily RiskMetrics value (J.P. Morgan/Reuters, 1996), which puts 84% of the weight on the last 30 days. The ES column reuses the unweighted ES.
- **Monte Carlo degrees of freedom.** $\nu = 6$ is fixed, not estimated. It equals the ν of the data process, which flatters the method. The scenarios use the sample covariance and mean, 100,000 draws, seed 7.
- **Backtest design.** The VaR is the historical VaR on the previous 500 days. The test uses the last 250 days of a separate 800-day path (same seed).
- **Stress inputs.** The 2008, March 2020 and 2022 shocks are hard-coded round numbers, and the hypothetical scenario is linear with default sensitivities (Section 7). There is no convexity and no second-round effect.
- **Ex-ante risk.** Sample covariance of the past 250 days, constant weights. The tracking error is only as good as that matrix.

4 Method

4.1 VaR and expected shortfall

Let $\alpha = 1\%$, $z = \Phi^{-1}(\alpha) < 0$, and let μ, σ, S, K be the sample mean, standard deviation, skewness and excess kurtosis of the portfolio returns.

Method	VaR	ES
Historical	$-q_\alpha(r)$	$-\text{mean}(r \mid r \le -\text{VaR})$
Age-weighted (BRW)	$-r_{(k)}$, first k with $\sum_{i\le k}\omega_{(i)} \ge \alpha$	same as historical
Normal	$-(\mu + z\sigma)$	$-\mu + \sigma\,\varphi(z)/\alpha$
Cornish-Fisher	$-(\mu + z_{cf}\,\sigma)$	$-(\mu + \sigma\,\overline{z_{cf}(u)})$, $u \in (0,\alpha)$
Monte Carlo (t)	$-q_\alpha$ of simulated returns	mean of simulated tail

Historical. The empirical 1% quantile of the portfolio returns (linear interpolation). Nothing is assumed about the shape, so it is as noisy as the 15 worst days behind it.

Age-weighted historical. Day i of n gets weight $\omega_i \propto (1-\lambda)\lambda^{n-i}$. Returns are sorted from worst to best and the weights are accumulated until they reach α. A larger λ gives a longer memory and a number closer to the plain historical VaR.

Normal. Driven by σ alone. Doubling the volatility doubles VaR and ES. The ES is always $\varphi(z)/(\alpha\,|z|) = 1.146$ times the VaR when $\mu = 0$.

Cornish-Fisher. The normal quantile is corrected for skewness and fat tails:

$$z_{cf} = z + \frac{(z^2-1)S}{6} + \frac{(z^3-3z)K}{24} - \frac{(2z^3-5z)S^2}{36}.$$

Positive skewness lowers the loss quantile and excess kurtosis raises it. The ES averages $z_{cf}(u)$ over 20,000 points of u between 10^{-6} and α.

Monte Carlo. With L the Cholesky factor of the sample covariance, $g \sim N(0, I)$ and $u \sim \chi^2_\nu/\nu$:

$$x = \mu + L\,\frac{g}{\sqrt{u}\,\sqrt{\nu/(\nu-2)}}, \qquad r_p = w^\top x.$$

The factor $\sqrt{\nu/(\nu-2)}$ keeps the sample covariance. A small u in one scenario inflates all five assets at once, which creates joint tail events. The degrees of freedom are fixed at $\nu = 6$, with 100,000 scenarios and seed 7.

4.2 Backtesting

The VaR is the historical VaR recalibrated every day on the previous 500 days. It is tested on the last 250 days. A breach is a loss larger than the VaR.

Test	Statistic	Distribution
Kupiec (POF)	$LR_{uc} = -2\ln\dfrac{(1-p)^{n-x}p^x}{(1-\hat{p})^{n-x}\hat{p}^x}$	χ^2_1
Christoffersen (1998, independence)	LR_{ind} from the transition counts n_{ij}	χ^2_1
Conditional coverage	$LR_{cc} = LR_{uc} + LR_{ind}$	χ^2_2
Basel traffic light	breaches in 250 days	green 0–4, yellow 5–9, red ≥ 10 breaches

Here n is the number of days, x the number of breaches, $p = 1\%$ and $\hat{p} = x/n$. The independence test compares a constant breach rate $\pi = (n_{01}+n_{11})/n$ with a Markov chain with rates $\pi_{01} = n_{01}/(n_{00}+n_{01})$ and $\pi_{11} = n_{11}/(n_{10}+n_{11})$. The traffic light adds 0.40 to 0.85 to the base multiplier of 3 in the yellow zone and 1.00 in the red zone.

4.3 Stress tests

- **Historical replay:** the P&L is $\sum_i w_i s_i$, where s_i is a fixed shock per asset for 2008, March 2020 and 2022.
- **Hypothetical:** asset P&L is $\beta_i \Delta E - D_i \Delta y - D_i^s \Delta s + x_i \Delta FX$. The scenario is equities -25%, rates $+150$ bp, spreads $+120$ bp and USD $+8\%$.

- **Correlation stress:** all off-diagonal correlations are set to 1, volatilities are kept, and the matrix is projected to the nearest positive semi-definite matrix. With $\rho = 1$ the portfolio volatility is the weighted sum of the asset volatilities, so there is no diversification at all.
- **Reverse stress:** find the shock x that minimises $x^\top \Sigma^{-1} x$ subject to $w^\top x = -L^*$ for a target loss L^* (SLSQP). The solution has the closed form $x^* = -L^* \Sigma w / (w^\top \Sigma w)$: each asset is shocked in proportion to its covariance with the portfolio, so the assets that drive the portfolio take most of the loss.

4.4 Ex-ante risk

With active weights $a = w_p - w_b$ and the annualised covariance Σ (daily covariance times 252):

$$\mathrm{TE}_{ex\text{-}ante} = \sqrt{a^\top \Sigma\, a}, \qquad \mathrm{VaR}_i = w_i \frac{-z\,(\Sigma w)_i}{\sqrt{w^\top \Sigma w}}.$$

The second formula is the Euler contribution of asset i to the normal VaR. The contributions add up to the total. The active weights are $+5$ points in each equity block and -5 points in sovereigns and gold.

5 Method Comparison

Method	Strength	Weakness
Historical	No distribution assumption	Rests on 15 days; one window, one number
Age-weighted (BRW)	Reacts within weeks	One day sets the VaR; ES is not weighted
Normal	Closed form, stable	Tail too thin; ES 13% under the historical one here
Cornish-Fisher	Uses skew and kurtosis, closed form	Unstable when K is large; ES integral extrapolates
Monte Carlo (t)	Joint scenarios, fits non-linear books	Needs ν and a covariance; sampling noise

For a linear book like this one, historical simulation is the benchmark everybody can audit, and Monte Carlo with a calibrated ν is the better tail estimate on a short sample. Cornish-Fisher is a quick check and should not stand alone when the sample kurtosis is high.

6 Results

VaR and ES, one day, 99% (portfolio of 1,500 days):

Method	VaR	ES
Historical	1.96%	2.38%
Age-weighted historical (BRW)	1.28%	2.38%
Parametric normal	1.81%	2.08%
Cornish-Fisher	2.15%	3.09%
Monte Carlo Student-t	2.00%	2.52%

The sample portfolio has a daily volatility of 0.79% (12.5% a year), skewness of +0.69 and excess kurtosis of 4.8. The normal model has the lowest ES because it ignores that kurtosis. Cornish-Fisher is highest: the kurtosis term adds −1.13 to the quantile, the skewness term gives back +0.51 and the squared-skewness term +0.18, which leaves $z_{cf} = -2.76$ against −2.33. The BRW VaR is lowest because it is a one-day figure: the worst return among the days that carry the weight is −1.28%, from the day nine sessions before the last one, and that single day holds 3.4% of the weight. The table does not say which method is right, because on one path there is no reference. Section 7 gives one: a 600,000-day run of the same process, where the historical VaR is 1.89% and the ES 2.64%.

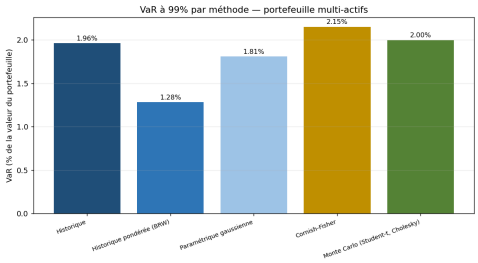


Figure 1: One-day 99% VaR by method (figure labels are in French).

Backtest (250 days, 500-day window): 4 breaches (1.6% against 1% expected).

Test	Statistic	p-value
Kupiec POF	0.7691	0.3805
Christoffersen independence	0.1306	0.7178
Conditional coverage	0.8998	0.6377

None of the tests rejects at 5%. The traffic light is green (cumulative binomial probability 89.22%, multiplier $3.00 + 0.00$). Four breaches against 2.5 expected is a normal outcome for the tests, and 89.22% is the chance of having four or fewer if the model is exact. The result does not show that the model is correct: repeating the same experiment on 200 other seeds gives a mean of 3.3 breaches, a Kupiec rejection in 24.5% of the samples (against 5% if the model were exact), and a green zone in 145 samples, yellow in 51 and red in 4. A one-year historical VaR on GARCH data probably breaches more often than nominal because volatility clusters, and seed 42 is not the typical draw: 4 breaches occur in 24 of the 200 samples, and the most frequent count is 1 (39 samples).

Stress tests (P&L in % of portfolio value):

Scenario	P&L
2008 replay	−27.65%
COVID, March 2020	−21.85%
2022 rate shock	−15.25%
Hypothetical (equities −25%, rates +150 bp, spreads +120 bp, USD +8%)	−16.60%

The ranking follows the size of the equity shock: 60% of the book takes −46% and −42% in 2008, while in 2022 bonds fall too (−16%), which removes the usual hedge and still gives only −15.25% because equities fall less. Forcing all correlations to 1 raises the annualised portfolio volatility from 12.54% to 15.95% (+27.2%). The least unlikely shock for a 15% loss (smallest

Mahalanobis distance) is mainly on equities: EU -20.65% and US -26.31%, with credit at -6.10%. Its Mahalanobis distance is 18.99, which is the loss of 15% divided by the daily portfolio volatility of 0.79%. Even this shock is 19 daily standard deviations away, so it is not a plausible day, and the covariance is a one-day covariance. The result gives the split of a large loss across assets, not a probability.

Ex-ante tracking error. The benchmark is 25% EU equities, 25% US equities, 25% government bonds, 15% credit and 10% gold. Calibrated on the first 250 days, the forecast TE is 2.13% against 1.68% realised over the next 250 days (ratio 0.79). Over 48 rolling windows (step 21 days), the realised TE is on average 1.17 times the forecast, with a median of 1.07 and a range of 0.62 to 2.26, and above 1 in 30 of the 48 windows. Raw beta is 1.173 and the Blume-style adjusted beta (Blume, 1975; the rounded rule $\frac{1}{3} + \frac{2}{3}\beta$) is 1.115. The information ratio is 0.602. A one-factor model with the benchmark as factor attributes 78.3% of the active variance to the factor and 21.7% to the specific part.

Asset class	Weight (%)	Share of normal VaR (%)
EU equities	30	41.3
US equities	30	52.6
Sovereign bonds	20	-0.35
Investment-grade credit	15	6.1
Gold	5	0.32

Equities are 60% of the capital and 94% of the VaR. Sovereign bonds are 20% of the capital and slightly reduce the risk, because their correlation with equities is -0.22 and -0.14. This is the diversification that the correlation stress removes. The forecast and realised TE disagree in both directions, so one window says little about the next year.

7 Worked Example

Normal VaR and ES. Sample mean $\mu = 0.0299\%$, $\sigma = 0.7900\%$, $z = -2.3263$, $\varphi(z) = 0.02665$:

$$\begin{aligned} \text{VaR} &= -(0.0299\% - 2.3263 \times 0.7900\%) = 1.8379\% - 0.0299\% = 1.81\%, \\ \text{ES} &= -0.0299\% + 0.7900\% \times \frac{0.02665}{0.01} = -0.0299\% + 2.1055\% = 2.08\%. \end{aligned}$$

Cornish-Fisher with the sample inputs $S = 0.694$, $K = 4.816$:

$$\begin{aligned} z_{cf} &= -2.3263 \ + 0.5107 \ - 1.1259 \ + 0.1815 = -2.7600, \\ \text{VaR}_{cf} &= -(0.0299\% - 2.7600 \times 0.7900\%) = 2.15\% \quad \text{against} \quad 1.81\% \text{ for the normal VaR.} \end{aligned}$$

Hypothetical scenario, asset by asset (the sensitivities are the package defaults):

$$\begin{aligned} \text{EU equities} &: 0.30 \times (-25\%) = -7.50\% \\ \text{US equities} &: 0.30 \times (-25\% + 1.0 \times 8\%) = 0.30 \times (-17\%) = -5.10\% \\ \text{Sovereign} &: 0.20 \times (-7 \times 0.0150) = 0.20 \times (-10.5\%) = -2.10\% \\ \text{Credit} &: 0.15 \times (-5 \times 0.0150 - 5 \times 0.0120) = 0.15 \times (-13.5\%) = -2.025\% \\ \text{Gold} &: 0.05 \times (0.3 \times 8\%) = 0.05 \times 2.4\% = +0.12\% \end{aligned}$$

The total is -16.605%, shown as -16.60% in the script output.

Kupiec test for the backtest above: $n = 250$, $x = 4$, $p = 0.01$, $\hat{p} = 0.016$, so $\ln L_0 = 246 \ln 0.99 + 4 \ln 0.01 = -20.8931$ and $\ln L_1 = 246 \ln 0.984 + 4 \ln 0.016 = -20.5085$:

$$LR_{uc} = -2\,(-20.8931 + 20.5085) = 0.7691 \quad \Rightarrow \quad p\text{-value} = 0.3805.$$

Sensitivity. All lines below rerun the package code.

- *Sampling noise.* The process was simulated for 200 seeds of 1,500 days. The 5th to 95th percentile of the 99% VaR (median in brackets) is 1.43–2.56% (1.77) for the historical method, 1.32–2.10% (1.55) for the normal, 1.47–2.32% (1.72) for Monte Carlo with 20,000 scenarios, 0.78–2.63% (1.33) for BRW and 1.54–3.95% (2.03) for Cornish-Fisher, whose maximum is 15.2% for a sample with excess kurtosis of 51.6. The normal VaR is below the historical one in 97.5% of the seeds and the normal ES in all of them. The gaps in the summary table are as large as the noise.
- *Long-run value.* A single 600,000-day run (seed 12345) gives excess kurtosis of 20.3, a historical VaR of 1.89% and ES of 2.64%. The normal model on that run gives 1.65% and 1.89%, 13% and 28% lower.
- *Monte Carlo ν.* With $\nu = 4$, 6, 10, 30 and the normal draw, VaR is 2.06, 2.00, 1.91, 1.83 and 1.80%, and ES 2.88, 2.52, 2.36, 2.15 and 2.07%. The ES moves three times as much as the VaR.
- *BRW decay.* $\lambda = 0.94$, 0.97, 0.98 all give 1.28%, because the same day sets the quantile. $\lambda = 0.99$, 0.995, 0.999 give 2.03, 2.39 and 2.07%.
- *Horizon.* For the historical VaR the square-root rule gives 4.38%, 6.20% and 8.77% at 5, 10 and 20 days. The VaR of overlapping sums of returns is 3.96%, 5.66% and 8.32%, so the rule overstates by 10.7%, 9.5% and 5.4% on this path.

8 Limits and Extensions

- **Synthetic data, linear positions:** the returns are simulated, the crisis shocks are approximations, and there is no option gamma or vega, so no conclusion applies to a real fund.
- **Normality in three places:** the parametric VaR, the Euler decomposition and the reverse stress test all assume normal returns.
- **Small samples:** with 250 days and about 2.5 expected breaches the tests have little power, and the tracking error ratios (0.79 and 1.17) come from one simulated path.

Three extensions would address these limits. Running the same code on real returns, with a volatility filter (GARCH or EWMA) applied before the historical quantile, would test whether the clustering seen in the breach counts disappears. Replacing the sample-moment Cornish-Fisher with a t or generalised Pareto fit to the tail would remove its dependence on a single kurtosis estimate. Linking the reverse stress to a multi-day covariance, or to a factor model, would turn the 19-sigma day into a shock with a horizon.

9 Conclusion

The package shows that a market-risk report is a set of numbers that depend on method, window and sample, and that a one-year backtest on a green light cannot separate a good model from a lucky one. A risk manager should ask for the range of the VaR across methods and seeds, not one figure, and should read the stress table as a map of exposure and not as a forecast.

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